

CBSE Class 10 Mathematics — Introduction to Trigonometry — Solutions

1. 1. If $\sin \theta = \frac{3}{5}$, then the value of $\cos \theta$ is:

Answer: (C)

Solution:

1. Use the identity $\sin^2 \theta + \cos^2 \theta = 1$
2. Substitute the value of $\sin \theta$
3. Solve for $\cos \theta$

Derivation:

$$\begin{aligned}\cos^2 \theta &= 1 - \sin^2 \theta = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25} \\ \implies \cos \theta &= \frac{4}{5}\end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

2. 2. The value of $\tan 45^\circ + \cot 45^\circ$ is:

Answer: (B)

Solution:

1. Recall the trigonometric values for standard angles.
2. Substitute $\tan 45^\circ = 1$ and $\cot 45^\circ = 1$.
3. Add the values.

Derivation:

$$\tan 45^\circ + \cot 45^\circ = 1 + 1 = 2$$

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3. 3. If $3 \tan \theta = 3$, then θ is equal to:

Answer: (C)

Solution:

1. Simplify the equation to find $\tan \theta$.
2. Identify the angle θ for which $\tan \theta = 1$.

Derivation:

$$\begin{aligned}3 \tan \theta &= 3 \\ \implies \tan \theta &= 1 \\ \implies \theta &= 45^\circ\end{aligned}$$

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4. 4. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is:

Answer: (A)

Solution:

1. Use the complementary angle identity $\cos(90^\circ - \theta) = \sin \theta$.
2. Substitute $\cos 72^\circ = \cos(90^\circ - 18^\circ) = \sin 18^\circ$.
3. Divide the terms.

Derivation:

$$\frac{\sin 18^\circ}{\cos 72^\circ} = \frac{\sin 18^\circ}{\sin 18^\circ} = 1$$

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5. 5. Which of the following is equal to $\sec^2 \theta - 1$?

Answer: (D)

Solution:

1. Recall the fundamental trigonometric identity $1 + \tan^2 \theta = \sec^2 \theta$.
2. Rearrange the identity to isolate $\sec^2 \theta - 1$.

Derivation:

$$\begin{aligned} 1 + \tan^2 \theta &= \sec^2 \theta \\ \implies \sec^2 \theta - 1 &= \tan^2 \theta \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

6. 6. Assertion (A): If $\sin \theta = \frac{1}{2}$, then $\theta = 30^\circ$. Reason (R): The value of $\sin \theta$ increases as θ increases from 0° to 90° .

Answer: (B)

Solution:

1. Check the validity of Assertion: For $\sin \theta = 1/2$, $\theta = 30^\circ$ is true.
2. Check the validity of Reason: The sine function is strictly increasing in the first quadrant, which is true.
3. Determine if R explains A: The fact that $\sin 30^\circ = 1/2$ is a specific value, its increasing nature is a general property, so R does not explain why $\sin 30^\circ$ is specifically $1/2$.

Derivation:

Assertion: $\sin 30^\circ = 0.5$

$$\begin{aligned} \implies \text{True. Reason: } \frac{d}{d\theta}(\sin \theta) &= \cos \theta > 0 \text{ for } \theta \in (0, 90^\circ) \\ &\implies \text{True.} \end{aligned}$$

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7. **7.** Assertion (A): The value of $\cos \theta$ can never be 1.5. Reason (R): The range of $\cos \theta$ is $[-1, 1]$.

Answer: (A)

Solution:

1. Check the validity of Assertion: Since the range of cosine is $[-1, 1]$, 1.5 is outside this range, so it is true.
2. Check the validity of Reason: By definition of trigonometric ratios in a right triangle, $\cos \theta = \text{base/hypotenuse} \leq 1$.
3. Determine if R explains A: The range definition is the direct reason why $\cos \theta$ cannot exceed 1.

Derivation:

$$-1 \leq \cos \theta \leq 1 \text{ for all } \theta \in \mathbb{R}$$

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8. **8.** Assertion (A): $\sin^2 \theta + \cos^2 \theta = 1$ for all θ . Reason (R): $\tan^2 \theta - \sec^2 \theta = 1$.

Answer: (C)

Solution:

1. Check the validity of Assertion: This is the fundamental Pythagorean identity, which is true.
2. Check the validity of Reason: The identity is $\sec^2 \theta - \tan^2 \theta = 1$. Therefore, $\tan^2 \theta - \sec^2 \theta = -1$. The reason is false.

Derivation:

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \text{ (Identity); } \sec^2 \theta - \tan^2 \theta = 1 \\ &\implies \tan^2 \theta - \sec^2 \theta = -1 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

9. **9.** Assertion (A): If $\tan A = \cot A$, then $A = 45^\circ$. Reason (R): $\tan A = \cot(90^\circ - A)$.

Answer: (A)

Solution:

1. Check the validity of Assertion: If $\tan A = \cot A$, then $\tan A = 1/\tan A$, so $\tan^2 A = 1$, $\tan A = 1$, $A = 45^\circ$. True.
2. Check the validity of Reason: This is the standard co-function identity.
3. Determine if R explains A: Yes, $\tan A = \cot A \implies \tan A = \tan(90^\circ - A) \implies A = 90^\circ - A \implies 2A = 90^\circ \implies A = 45^\circ$.

Derivation:

$$\begin{aligned} \tan A &= \cot A \\ \iff \tan A &= \tan(90^\circ - A) \\ \iff A &= 90^\circ - A \\ \iff A &= 45^\circ \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

10. **10.** Assertion (A): The value of $\sin 0^\circ$ is 0. Reason (R): The value of $\cos 0^\circ$ is 0.

Answer: (C)

Solution:

1. Check the validity of Assertion: $\sin 0^\circ = 0$. True.
2. Check the validity of Reason: $\cos 0^\circ = 1$. False.

Derivation:

$$\sin 0^\circ = 0, \cos 0^\circ = 1$$

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11. **11.** If $\tan A = \frac{4}{3}$, find the value of $\sin A + \cos A$.

Answer:

$$\frac{7}{5}$$

Solution:

1. Use the definition of $\tan A = \frac{\text{opposite}}{\text{adjacent}}$ to find the hypotenuse using the Pythagorean theorem.
2. Calculate $\sin A$ and $\cos A$ and find their sum.

Derivation:

$$\tan A = \frac{BC}{AB}$$

$$\Rightarrow BC = 4k, AB = 3k. AC = \sqrt{(4k)^2 + (3k)^2} = 5k. \sin A = \frac{4}{5}, \cos A = \frac{3}{5}. \sin A + \cos A = \frac{4}{5} + \frac{3}{5}$$

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12. **12.** Evaluate: $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{1}$

Answer:

$$2$$

Solution:

1. Substitute the standard trigonometric values for 45° , 30° , and 60° .
2. Simplify the expression using arithmetic operations.

Derivation:

$$\tan 45^\circ = 1, \cos 30^\circ = \frac{\sqrt{3}}{2}, \sin 60^\circ = \frac{\sqrt{3}}{2}. \text{Expression} = 2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

13. **13.** If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .

Answer:

$$A = 45^\circ, B = 15^\circ$$

Solution:

1. Use the inverse trigonometric values to set up a system of linear equations.
2. Solve the equations for A and B .

Derivation:

$$\begin{aligned} A - B = 30^\circ \text{ and } A + B = 60^\circ. & \quad (A - B) + (A + B) = 30^\circ + 60^\circ \\ & \implies 2A = 90^\circ \\ & \implies A = 45^\circ. B = 60^\circ - 45^\circ = 15^\circ \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

14. 14. Express $\cos A$ in terms of $\tan A$.

Answer:

$$\cos A = \frac{1}{\sqrt{1 + \tan^2 A}}$$

Solution:

1. Use the identity $\sec^2 A = 1 + \tan^2 A$.
2. Relate $\sec A$ to $\cos A$ using the reciprocal relation.

Derivation:

$$\begin{aligned} \sec^2 A &= 1 + \tan^2 A \\ \implies \sec A &= \sqrt{1 + \tan^2 A}. \text{ Since } \cos A = \frac{1}{\sec A}, \cos A = \frac{1}{\sqrt{1 + \tan^2 A}} \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

15. 15. Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

Answer:

$$1$$

Solution:

1. Apply the complementary angle relation $\tan \theta = \cot(90^\circ - \theta)$.
2. Simplify the fraction.

Derivation:

$$\tan 65^\circ = \tan(90^\circ - 25^\circ) = \cot 25^\circ. \text{ Expression} = \frac{\cot 25^\circ}{\cot 25^\circ} = 1$$

Introduction to Trigonometry — Trigonometric Ratios of Complementary Angles

16. 16. If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .

Answer:

$$A = 45^\circ, B = 15^\circ$$

Solution:

1. Use the values of tangent from the trigonometric table for $A + B$ and $A - B$.

2. Set up a system of linear equations in A and B .
3. Solve the system using elimination method.

Derivation:

$$\begin{aligned}
 \tan(A + B) &= \sqrt{3} \\
 \implies A + B &= 60^\circ \quad (1) \\
 \tan(A - B) &= \frac{1}{\sqrt{3}} \\
 \implies A - B &= 30^\circ \quad (2) \\
 (1) + (2) : 2A &= 90^\circ \\
 \implies A &= 45^\circ \\
 (1) - (2) : 2B &= 30^\circ \\
 \implies B &= 15^\circ
 \end{aligned}$$

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17. 17. If $3 \cot \theta = 2$, evaluate the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.

Answer:

$$\frac{1}{3}$$

Solution:

1. Given $3 \cot \theta = 2$, find $\cot \theta = \frac{2}{3}$.
2. Divide numerator and denominator by $\sin \theta$ to express in terms of $\cot \theta$.
3. Substitute the value of $\cot \theta$ and simplify.

Derivation:

$$\begin{aligned}
 \frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta} &= \frac{4 - 3 \cot \theta}{2 + 6 \cot \theta} \\
 &= \frac{4 - 3(2/3)}{2 + 6(2/3)} \\
 &= \frac{4 - 2}{2 + 4} = \frac{2}{6} = \frac{1}{3}
 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

18. 18. Prove that $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

Answer:

Proof completed.

Solution:

1. Expand both squares using $(a + b)^2 = a^2 + b^2 + 2ab$.
2. Group $\sin^2 A + \cos^2 A = 1$ and use identities $\csc^2 A = 1 + \cot^2 A$ and $\sec^2 A = 1 + \tan^2 A$.
3. Simplify the product terms $2 \sin A \csc A$ and $2 \cos A \sec A$ to 2.

Derivation:

$$\begin{aligned} LHS &= \sin^2 A + \csc^2 A + 2 + \cos^2 A + \sec^2 A + 2 \\ &= (\sin^2 A + \cos^2 A) + 4 + (1 + \cot^2 A) + (1 + \tan^2 A) \\ &= 1 + 4 + 1 + 1 + \tan^2 A + \cot^2 A = 7 + \tan^2 A + \cot^2 A = RHS \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

19. 19. If $\sin \theta + \sin^2 \theta = 1$, prove that $\cos^2 \theta + \cos^4 \theta = 1$.

Answer:

Proof completed.

Solution:

1. From $\sin \theta + \sin^2 \theta = 1$, express $\sin \theta = 1 - \sin^2 \theta$.
2. Substitute $\sin \theta = \cos^2 \theta$.
3. Square both sides to get $\sin^2 \theta = \cos^4 \theta$ and substitute into the identity.

Derivation:

$$\begin{aligned} \sin \theta &= 1 - \sin^2 \theta \\ \implies \sin \theta &= \cos^2 \theta \\ \text{Square both sides: } \sin^2 \theta &= \cos^4 \theta \\ \text{Substitute into } \cos^2 \theta + \cos^4 \theta : \sin \theta + \sin^2 \theta &= 1 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities

20. 20. Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$.

Answer:

$$\frac{3\sqrt{2} - \sqrt{6}}{8}$$

Solution:

1. Substitute the standard trigonometric values: $\cos 45^\circ = 1/\sqrt{2}$, $\sec 30^\circ = 2/\sqrt{3}$, $\csc 30^\circ = 2$.
2. Simplify the denominator.
3. Rationalize the resulting expression.

Derivation:

$$\begin{aligned} \frac{1/\sqrt{2}}{2/\sqrt{3} + 2} &= \frac{1/\sqrt{2}}{(2 + 2\sqrt{3})/\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{2}(2 + 2\sqrt{3})} \\ &= \frac{\sqrt{3}}{2\sqrt{2} + 2\sqrt{6}} = \frac{\sqrt{3}(2\sqrt{6} - 2\sqrt{2})}{4(6 - 2)} = \frac{6\sqrt{2} - 2\sqrt{6}}{16} = \frac{3\sqrt{2} - \sqrt{6}}{8} \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios of Specific Angles

21. 21. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$.

Answer:

$$\csc A + \cot A$$

Solution:

1. Divide numerator and denominator by $\sin A$.
2. Rewrite the expression in terms of $\cot A$ and $\csc A$.
3. Replace 1 in the numerator with $\csc^2 A - \cot^2 A$.
4. Factorize the numerator using $a^2 - b^2 = (a - b)(a + b)$.
5. Cancel the common term to simplify.

Derivation:

$$\frac{\cot A - 1 + \csc A}{\cot A + 1 - \csc A} = \frac{\cot A + \csc A - (\csc^2 A - \cot^2 A)}{\cot A - \csc A + 1} = \frac{(\cot A + \csc A) - (\csc A - \cot A)(\csc A + \cot A)}{\cot A - \csc A + 1}$$

Introduction to Trigonometry — Trigonometric Identities

- 22. 22.** If $\sin \theta + \cos \theta = \sqrt{3}$, then prove that $\tan \theta + \cot \theta = 1$.

Answer:

$$\tan \theta + \cot \theta = 1$$

Solution:

1. Square both sides of the given equation.
2. Expand $(\sin \theta + \cos \theta)^2$ to get $\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 3$.
3. Use identity $\sin^2 \theta + \cos^2 \theta = 1$ to find $2 \sin \theta \cos \theta = 2$, so $\sin \theta \cos \theta = 1$.
4. Express $\tan \theta + \cot \theta$ as $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$.
5. Simplify the fraction to $\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{1} = 1$.

Derivation:

$$\begin{aligned} (\sin \theta + \cos \theta)^2 &= 3 \\ \implies 1 + 2 \sin \theta \cos \theta &= 3 \\ \implies \sin \theta \cos \theta &= 1; \quad \tan \theta + \cot \theta = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} = 1 \end{aligned}$$

Introduction to Trigonometry — Trigonometric Ratios

- 23. 23.** Evaluate: $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

Answer:

$$\frac{67}{12}$$

Solution:

1. State values: $\cos 60^\circ = 1/2$, $\sec 30^\circ = 2/\sqrt{3}$, $\tan 45^\circ = 1$, $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$.
2. Substitute values into the expression.
3. Calculate squares: $\cos^2 60^\circ = 1/4$, $\sec^2 30^\circ = 4/3$, $\tan^2 45^\circ = 1$.
4. Simplify the numerator: $5(1/4) + 4(4/3) - 1$.
5. Simplify the denominator: $(1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 = 1$.
6. Final calculation: $5/4 + 16/3 - 1 = (15 + 64 - 12)/12 = 67/12$.

Derivation:

$$\frac{5(1/4) + 4(4/3) - 1}{1} = \frac{5}{4} + \frac{16}{3} - 1 = \frac{15 + 64 - 12}{12} = \frac{67}{12}$$

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24. 24. If $\sec \theta = x + \frac{1}{4x}$, prove that $\sec \theta + \tan \theta = 2x$ or $\frac{1}{2x}$.

Answer:

$$\sec \theta + \tan \theta = 2x \text{ or } \frac{1}{2x}$$

Solution:

1. Use the identity $\tan^2 \theta = \sec^2 \theta - 1$.
2. Substitute $\sec \theta = x + \frac{1}{4x}$.
3. Calculate $\tan^2 \theta = (x + \frac{1}{4x})^2 - 1$.
4. Simplify to $\tan^2 \theta = x^2 + \frac{1}{16x^2} + \frac{1}{2} - 1 = x^2 + \frac{1}{16x^2} - \frac{1}{2}$.
5. Notice this is a perfect square: $(x - \frac{1}{4x})^2$.
6. Thus $\tan \theta = \pm(x - \frac{1}{4x})$.
7. Sum $\sec \theta + \tan \theta$ for both cases.

Derivation:

$$\begin{aligned} \tan^2 \theta &= (x + \frac{1}{4x})^2 - 1 = x^2 + \frac{1}{16x^2} - \frac{1}{2} = (x - \frac{1}{4x})^2 \\ \implies \tan \theta &= \pm(x - \frac{1}{4x}); \quad \text{Case 1: } (x + \frac{1}{4x}) + (x - \frac{1}{4x}) = 2x; \quad \text{Case 2: } (x + \frac{1}{4x}) - (x - \frac{1}{4x}) = \frac{1}{2x} \end{aligned}$$

Introduction to Trigonometry — Trigonometric Identities